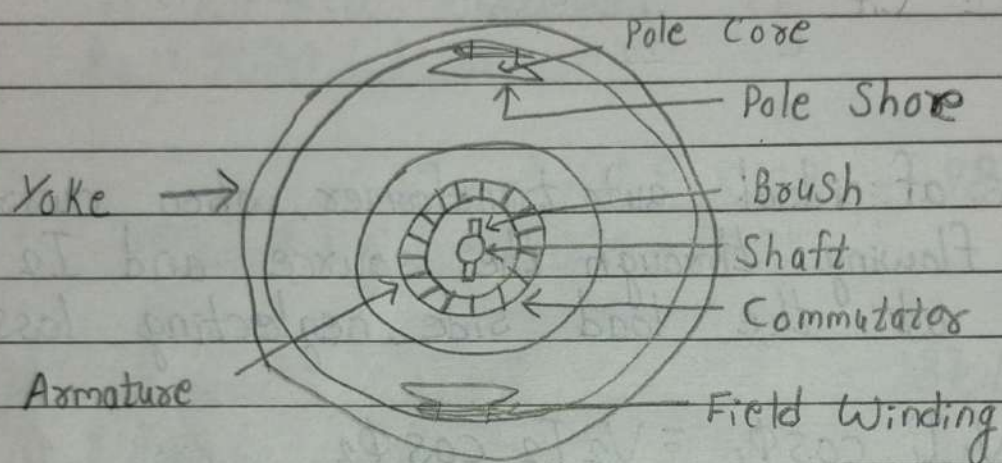


MODULE - 4

Electrical Machines

Imp. # Parts of DC Machine



The main parts of a DC machine are:

- ① Yoke: The outer frame that provides mechanical support and protects from the atmospheric ~~air~~ dust.
- ② Poles: Electromagnets that are attached to the yoke, with each pole having a pole core and a pole shoe.
- ③ Field Windings: Coils of wire wound around the pole cores to create the magnetic field.

- ④ Brushes: Stationary carbon blocks that collect current from the commutator.
- ⑤ Commutator: A segmented copper cylinder that converts the AC in the armature winding to DC.
- ⑥ Armature Core: A laminated cylinder with slots that houses the armature winding.
- ⑦ Armature Winding: Coils of wire placed in the armature core slots where voltage is induced.
- ⑧ Shaft: The rotating shaft that provides the mechanical output.

A DC machine works on the principle of electro-magnetism, converting mechanical to electrical energy (generator) or vice-versa (motor).

EMF Equation of DC Generator

When supply is given to the armature coil then an emf is induced in the armature coil.

Let, P = No. of poles

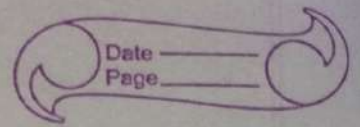
Φ = Flux per pole (in Wb)

N = Speed (in RPM)

Z = Total no. of conductor

A = No. of parallel path in armature.

Wave Winding $\Rightarrow A=2$
Lap Winding $\Rightarrow A=P$



Z/A = Total no. of conductors per path.

Since, flux per pole (Φ) cuts the rotor conductors in one rotation, then average voltage in one rotation is given by;

$$E_{avg} = \frac{\text{Flux cut in one rotation } (P\Phi)}{\text{Time taken in one rotation}}$$

'n' rotation in 1 second.

1 rotation in $1/n$ second.

$$\text{So, } E_{avg} = \frac{P\Phi}{1/n}$$

$$\Rightarrow E_{avg} = P\Phi n$$

$$\Rightarrow E_{avg} = \frac{P\Phi N}{60}$$

$$\Rightarrow E = E_{avg} \times \frac{Z}{A} \quad [\text{For Total Voltage}]$$

$$\therefore E = \frac{P\Phi N Z}{60A}$$

- 9) A four pole, wave winding has 720 conductors and it is rotated 1000 RPM. If the useful flux is 20 mWb. Calculate generated voltage.

Solⁿ:

$$P = 4$$

$$Z = 720$$

$$N = 1000 \text{ RPM}$$

$$\phi = 20 \text{ mWb} = 20 \times 10^{-3} \text{ Wb}$$

$$A = 2$$

We know,

$$E = \frac{P\phi NZ}{60A} = \frac{4 \times 20 \times 10^{-3} \times 1000 \times 720}{60 \times 2} = 480 \text{ V.}$$

- Q) A 8 pole, lap winding has 40 slots with 12 conductor per slot, generates a voltage of 500 V. Determine the speed at which it is running if flux per pole is 50 mWb.

Solⁿ:

$$P = 8$$

$$A = P = 8$$

$$Z = 40 \times 12 = 480$$

$$E = 500 \text{ V}$$

$$N = ?$$

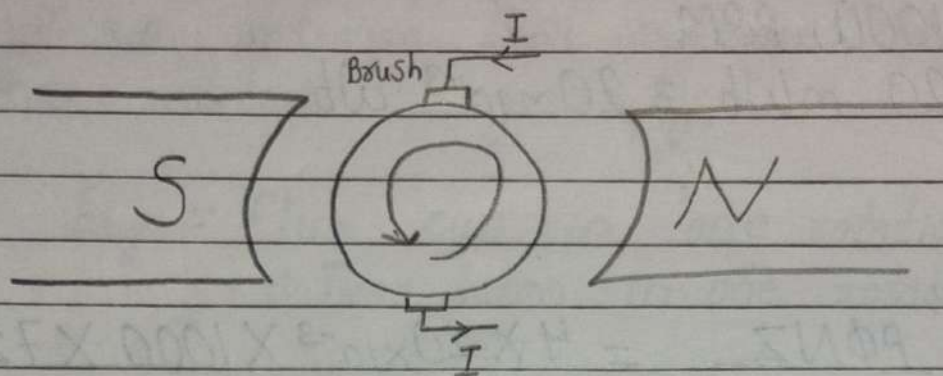
$$\phi = 50 \text{ mWb} = 50 \times 10^{-3} \text{ Wb}$$

We have,

$$E = \frac{P\phi NZ}{60A}$$

$$\Rightarrow N = \frac{60EA}{P\phi Z} = \frac{60 \times 500 \times 8}{8 \times 50 \times 10^{-3} \times 480} = 1,250 \text{ RPM.}$$

Principle Operation of DC Motor



A DC motor operates on the principle that, when a current-carrying conductor is placed in the magnetic field then it experiences a mechanical force (or Lorentz Force) and explained by left-hand rule.

This force generates a torque that causes the motor's armature to rotate, converting electrical energy into mechanical energy.

Torque Equation of DC Motor

The mechanical power required to rotate the shaft is given by,

$$P_{\text{mech.}} = \tau \omega = \tau \frac{2\pi N}{60} \quad \text{--- (1)}$$

Also, the mechanical power produced by the motor on electrical side is,

$$P_{\text{mech.}} = E_b I_a \quad \text{--- (2)}$$

where, E_b is back emf
& I_a is armature current.

From eq^{ns}. (1) & (2),

$$E_b I_a = \tau \frac{2\pi N}{60}$$

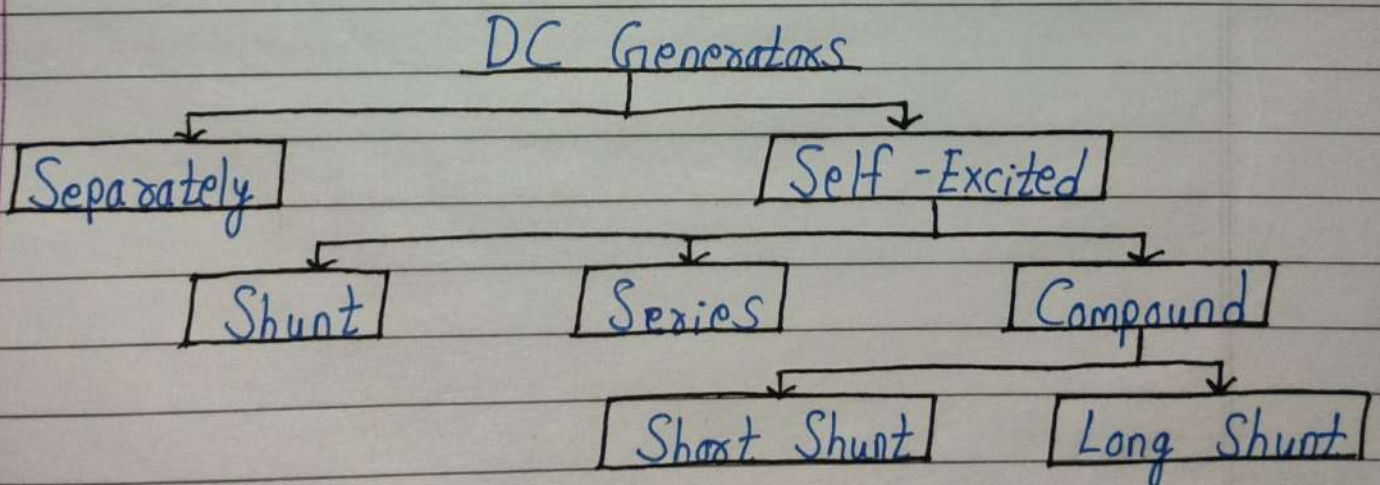
$$\propto, \frac{N\Phi ZP}{60A} I_a = \tau \frac{2\pi N}{60}$$

$$\therefore \tau = \frac{P\Phi Z I_a}{2\pi A}$$

$$\Rightarrow \tau \propto \Phi I_a$$

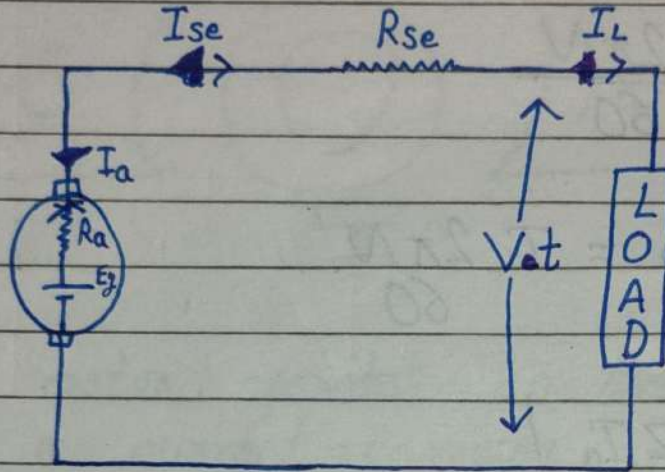
Thus, the torque produced by the DC motor is proportional to the flux and armature current.

Types of DC Generators



1. Series D.C. Generator

Field winding is connected in series with the armature winding and the external load.



$$V_t - I_{se}R_{se} - I_aR_a - E_g = 0$$

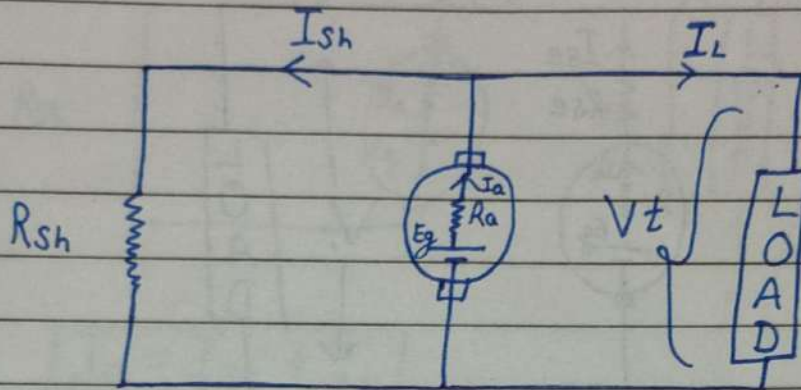
$$V_t = I_{se}R_{se} + I_aR_a + E_g$$

$$\therefore \boxed{V_t = I_a(R_{se} + R_a) + E_g} \quad [\because I_{se} = I_a]$$

Imp.
2.

Shunt D.C. Generator

Field winding is connected in parallel with the armature winding.



$$I_a = I_{sh} + I_L$$

$$E_g - I_a R_a - V_t = 0$$

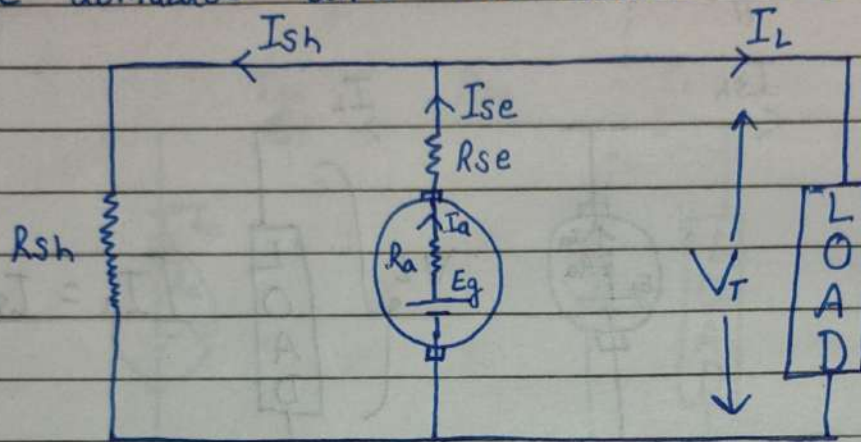
$$V_t = E_g - I_a R_a$$

Brush drop,

$$E_g - I_a R_a - V_t - V_{brush} = 0$$

3. Long Shunt D.C. Generator

Shunt field winding is connected in parallel with both the armature and the series field windings.



$$I_a = I_{se} = I_L + I_{sh}$$

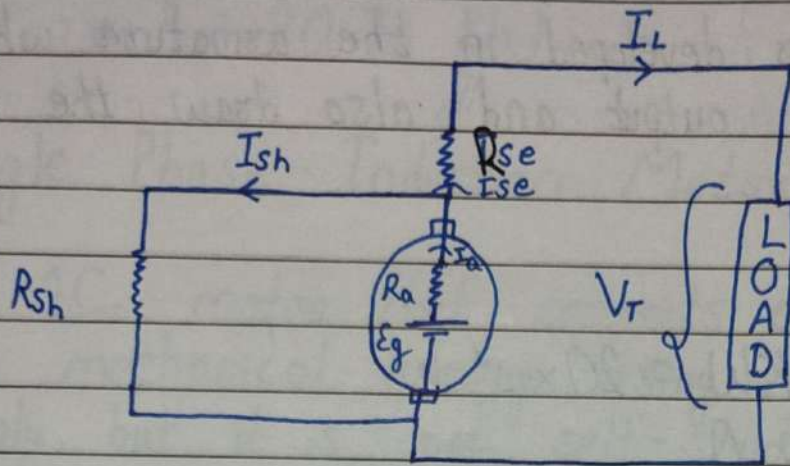
$$E_g - I_a R_a - I_a R_{se} - V_T = 0$$

$$E_g - I_a (R_a + R_{se}) = V_T$$

$$P_g = I_a E_g$$

$$P_L = V_T I_L$$

4. Short Shunt D.C. Generator



$$I_a = I_{sh} + I_L$$

$$\begin{aligned} \cancel{I_{sh}} \Rightarrow E_g - I_a R_a - I_L R_{sh} - V_t &= 0 \\ \Rightarrow V_t &= E_g - I_a R_a - I_L R_{sh} \end{aligned}$$

$$I_{sh} = \frac{V_t}{R_{sh}}$$

Q) 20 kW, 200 V DC shunt generator has an armature resistance of 0.05Ω , shunt resistance 200Ω . Calculate power developed in the armature when it delivers rated output and also draw the circuit diagram.

Solⁿ:

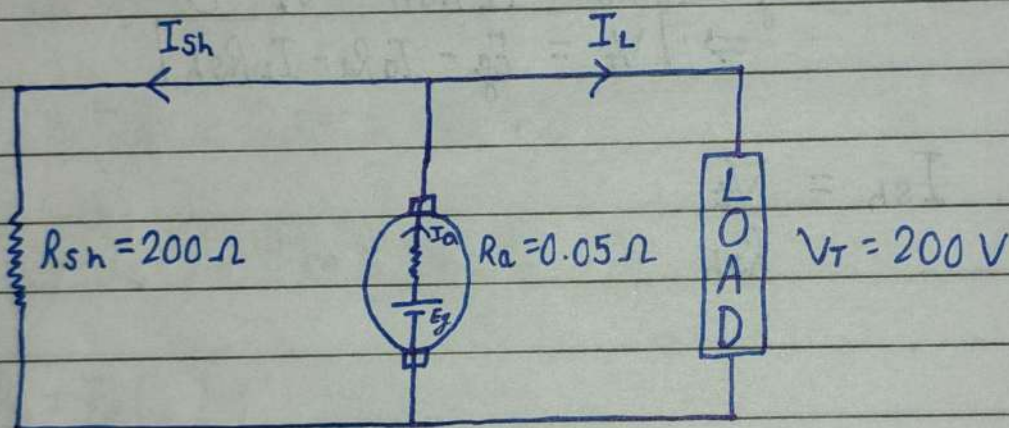
$$P_L = 20 \text{ kW} = 20 \times 10^3 \text{ W}$$

$$R_a = 0.05 \Omega$$

$$V_T = 200 \text{ V}$$

$$R_{sh} = 200 \Omega$$

$$P_g = P_a = ?$$



$$I_L = \frac{P_L}{V_T} = \frac{20 \times 10^3}{200} = 100 \text{ A}$$

$$I_{sh} = \frac{V_T}{R_{sh}} = \frac{200}{200} = 1 \text{ A}$$

$$I_a = I_L + I_{sh} = 101 \text{ A}$$

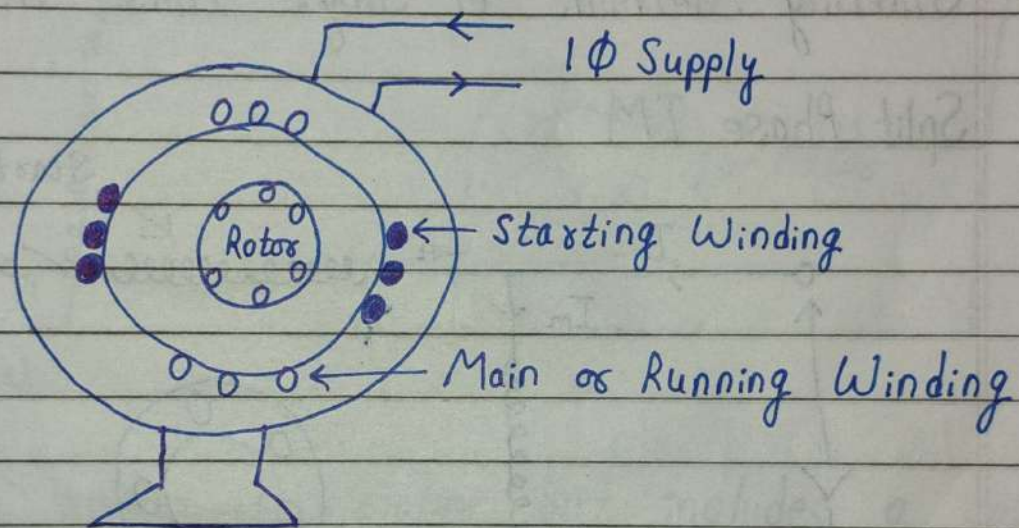
$$E_g = V_T + I_a R_a = 200 + 101 \times 0.05 = 205.05 \text{ V}$$

$$\therefore P_a = E_g \times I_a = 205.05 \times 101 = 20,710.05 \text{ W}$$

$$\text{i.e. } P_a = 20.71 \text{ kW}$$

Single Phase Induction Motor

An A.C. motor that converts electrical energy into mechanical energy using a single-phase power supply, but it is not self-starting.



When a single-phase A.C. supply is connected to the stator's main winding, it creates a pulsating magnetic field.

This pulsating field can be seen as two equal magnetic fields rotating in opposite directions at synchronous speed.

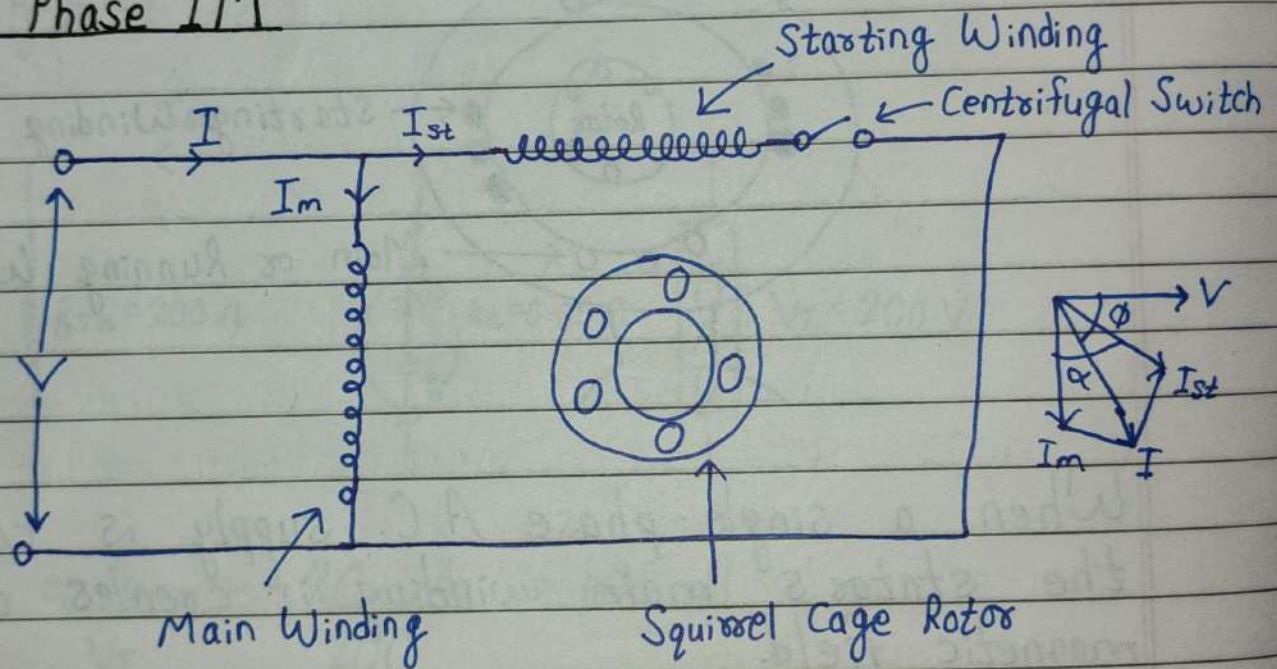
These two fields create equal and opposite torques, so the motor has no starting torque (net torque is zero) and cannot start by itself.

To start the motor, an auxiliary (starting) winding is added along with a capacitor in series. This creates a phase difference between the currents in the two windings, producing a rotating magnetic field that starts the rotation.

Once the motor starts, the rotating field induces current in the rotor conductors.

Starting Methods of Single Phase Induction Motors

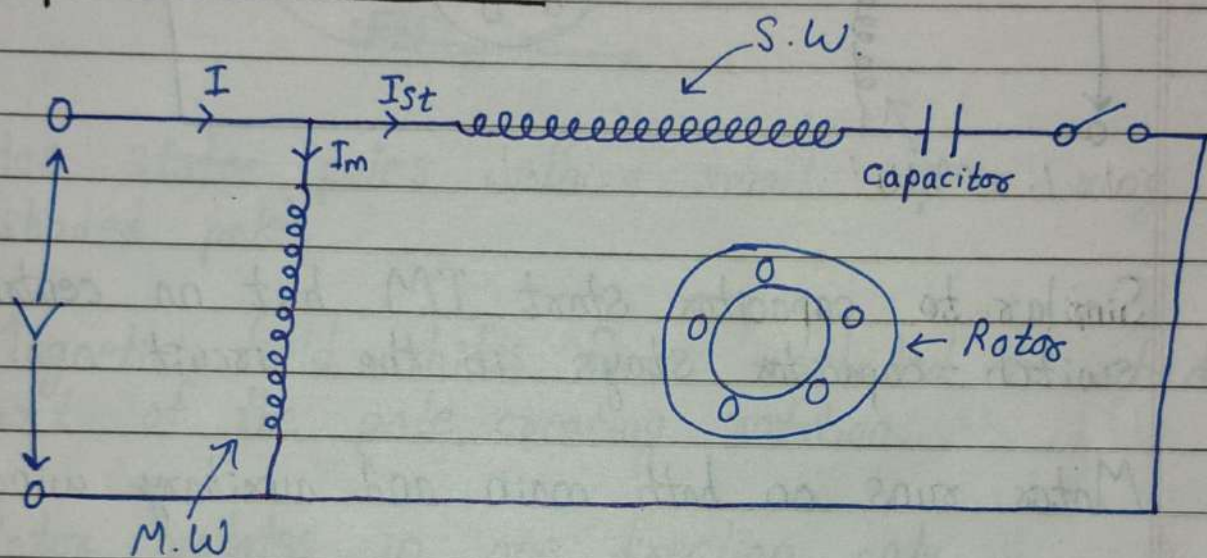
1. Split Phase IM



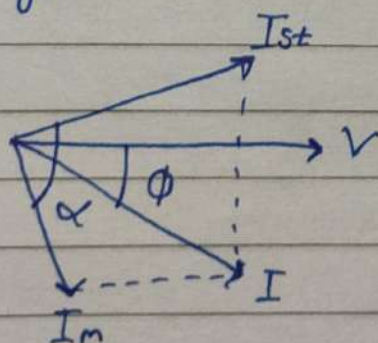
- Has two windings; main & starting (auxiliary) winding.
- Auxiliary winding has high resistance while main winding is inductive.
- Two currents (I_m & I_{st}) produce fluxes that create a rotating magnetic field.

- A centrifugal switch disconnects the auxiliary winding at 70-80% of synchronous speed.
- Main winding runs the motor continuously.

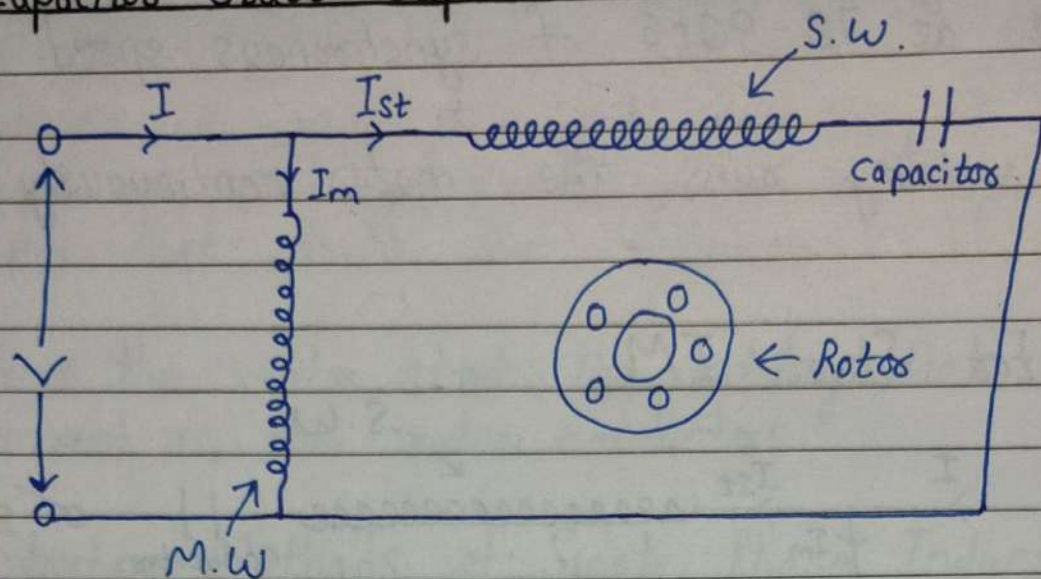
2. Capacitor Start IM



- Similar to split-phase motor but includes a capacitor in series with the starting winding.
- Capacitor (in series) improves phase difference, giving high starting torque.
- Known for better starting performance than split-phase type.

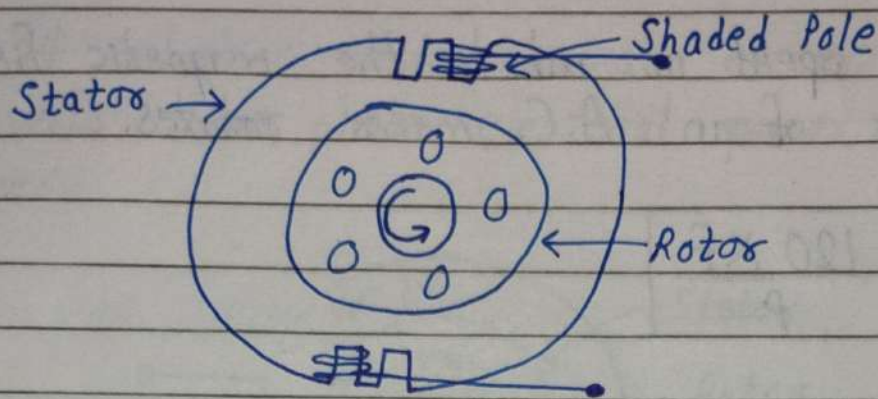


3. Capacitor Start Capacitor Run IM



- Similar to capacitor start IM but no centrifugal switch - capacitor stays in the circuit.
- Motor runs on both main and auxiliary windings.
- Improves power factor and efficiency during running.
- Used for smooth operation and high efficiency.

4. Shaded Pole IM



- Has stator poles with a small copper ring (shaded pole).
- Magnetic field shifts from unshaded to shaded part of the pole, creating rotation.
- Rotor rotates in one direction only.
- Common in fans, hair dryers and small appliances.
- Simple, low cost, but low starting torque.

Synchronous Speed

It is the speed at which the magnetic field in the stator of an A.C. motor rotates.

$$N_s = \frac{120 \times f}{P}$$

where, $f \rightarrow$ Supply Frequency
& $P \rightarrow$ No. of Poles

Slip

It shows how much slower the rotor turns compared to the rotating magnetic field.

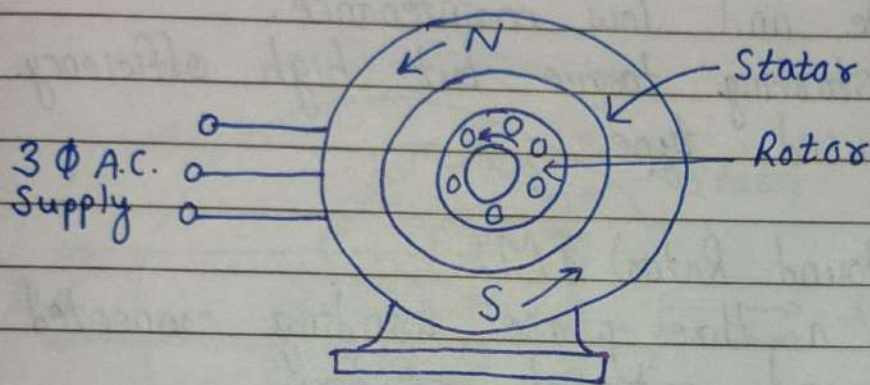
Slip is the difference between the synchronous speed and the actual rotor speed of an induction motor.

$$S = \frac{N_s - N_r}{N_s} \times 100$$

where, $N_s \rightarrow$ Synchronous Speed (RPM)
& N_r or $N_r \rightarrow$ Rotor Speed (RPM)

Three Phase Induction Motor

A 3ϕ induction motor is an electric motor that converts three phase electrical power into mechanical power.



- When a 3ϕ A.C. supply is given to the stator, it creates a rotating magnetic field.
- This rotating field cuts the rotor conductors and induces an emf.
- The induced emf causes current to flow in the rotor conductors.
- The interaction between the rotating magnetic field and rotor currents produces torque, which makes the rotor turn in the same direction as the rotating field.
- The rotor speed is slightly less than the synchronous speed.

Types of 3ϕ IM

1. Squirrel Cage IM:

- Rotor bars are short-circuited by end rings (like a cage).
- Rugged, simple and low maintenance.
- Has low starting torque but high efficiency.
- Commonly used type.

2. Slip Ring (Wound Rotor) IM:

- Rotor has a three-phase winding connected to slip rings and external resistors.
- The external resistance helps control starting torque and speed.
- Used where high starting torque or speed control is needed.

Applications of 3ϕ IM

- Fans, pumps, compressors
- Domestic and industrial drives
- Cranes, elevators, hoists.
- Mills, crushers, large compressors.

Synchronous Machine

A synchronous machine ~~works on~~ is an electrical machine that operates at a constant speed, known as synchronous speed.

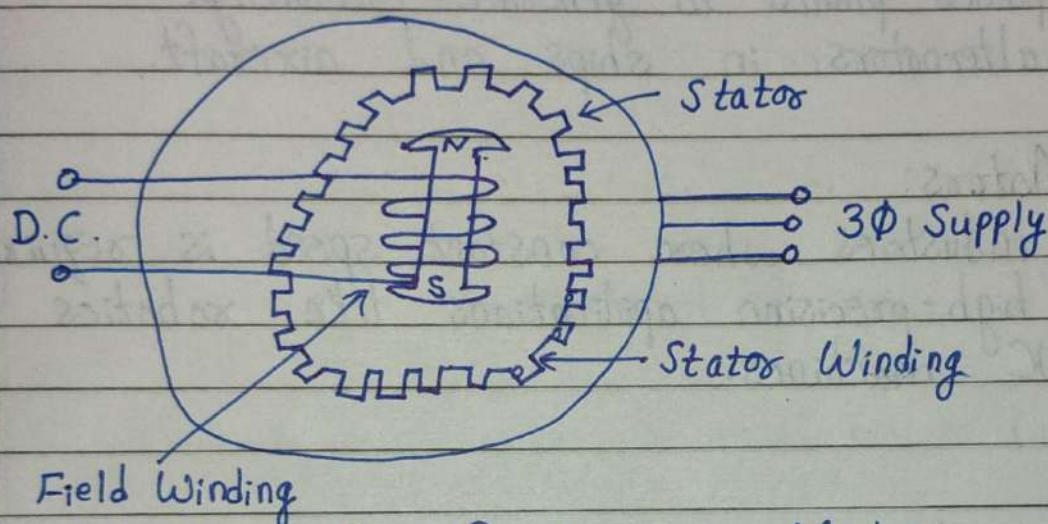


Fig: Synchronous Motor

A synchronous machine works on the principle of electromagnetic induction and magnetic locking between the stator and rotor magnetic fields.

Stator has a 3Φ winding connected to an AC supply. This produces a rotating magnetic field at synchronous speed.

Rotor carries a DC-excited winding (or permanent magnets) that creates a steady magnetic field.

When the rotor is brought up to synchronous speed, its magnetic field locks with the stator's rotating field.

Once locked, both rotate together at the same speed, there is no slip.

Applications of Synchronous Machine

As Generators:

- In power plants to generate electricity.
- As alternators in ships and aircraft.

As Motors:

- In industries where constant speed is required.
- In high-precision applications like robotics and CNC machines.

Q) Why single phase induction motor is not self-starting?

⇒ A single phase induction motor is not self-starting because a single phase AC supply produces a pulsating magnetic field, not a rotating one. This creates equal and opposite torques resulting in zero starting torque.